

THE LITTLE BROADWAY · MOBILE APP

End-to-End Testing Report

API integration · UI/UX & design · robustness & edge cases · static analysis ·
automated test suite

0

Analyzer errors

130/130

Tests passing

17

High findings

16

Medium findings

Date: 17 June 2026

Build: dev-vishesh · Flutter 3.38

1. Executive Summary

The TLB mobile app is in a healthy, near-release state: it compiles with zero analyzer errors, the full automated suite (130 tests) passes, and core infrastructure - secure token storage, network timeouts, lifecycle disposal, the real Razorpay payment path, and image error-fallbacks - is well-engineered. This end-to-end pass exercised the API layer, every screen/widget/section, design-system consistency, accessibility, and edge-case robustness.

The findings below are overwhelmingly hardening items rather than broken features. The dominant themes are: (a) defensive JSON parsing - several required casts and decodes can throw on unexpected payloads, most critically in the booking/payment models; (b) network resilience - there is no centralized 401-refresh-retry; (c) live-data readiness - home sections still render mock data with their empty-state guards commented out; and (d) design-system drift - brand colors are hardcoded across many files instead of pulled from AppColors. None of these block day-to-day use today, but items in Sections 4-6 should be addressed before wiring live API data and shipping.

Verdict: SHIP-READY for the current mock/demo build. Address the Section 7 priority list (crash-safety in the money flow + network resilience) before enabling live API data.

2. Automated Test & Analysis Results

- flutter analyze - 0 errors, 0 warnings; 29 info-level style lints (below).
- flutter test - 130 / 130 passing (widget, unit & smoke tests). No RenderFlex overflow surfaced.

Static-analysis lints (cosmetic, non-blocking):

Lint rule	Count	Location
curly_braces_in_flow_control_structures	1	booking_detail_screen.dart:1615
use_null_aware_elements (prefer ?. spread)	28	classes/events/programs_listing_service.dart

3. API Endpoint Inventory

All ~40 REST endpoints reach the configured host (<https://tlb-api.reluconsultancy.in/api/v1>). Grouped by service:

Service	Methods	Paths
Auth	POST	/auth/request-otp/ · /auth/verify-otp/
Auth	POST	/auth/google-login/ · /auth/refresh-token/ · /auth/logout/
Auth	POST	/auth/customer/password/reset/{request,verify-otp,confirm}/
Auth	GET/PATCH/DELETE	/customer/profile/ · /customer/account/
Wishlist	GET/POST/DELETE	/wishlist/ · /wishlist/add/ · /wishlist/remove/{id}/
Reviews	GET/POST/PATCH/DELETE	/listings/{id}/reviews/ · /reviews/{id}/ · /customer/reviews/
Reviews	GET/POST/DELETE	/reviews/{id}/media/ · /listings/{id}/my-review/
Partner	GET/POST/DELETE	/partner/{id}/follow/ · /unfollow/ · /partner/followed/
Notifications	GET/POST	/notifications/in-app/ · /unread-count/ · /{id}/read/ · /read-all/
Home feed	GET	/homepage/sections/

Service	Methods	Paths
Listings	GET	/events · /classes · /programs · /venues (+ detail, paginated)
Bookings	POST	/bookings/initiate/ · /initiate-with-saved-method/ · /{id}/verify-payment/
Bookings	GET/POST	/bookings/ · /{id}/ · /{id}/ticket/data/ · /{id}/cancel/

4. Findings Overview

17

High severity

16

Medium severity

3

Low severity

36

Total findings

Severity = High: can crash or corrupt the money/auth flow, or blocks live data. Medium: degraded UX, robustness or consistency risk. Low: polish / minor hardening.

5. API & Networking Findings

HIGH Unguarded jsonDecode(...) as Map on protected calls

Where booking_service.dart:69,140,185,... · review_service.dart (no try/catch) · partner_service.dart:89

Issue A 500 HTML page, empty body, or bare JSON array makes jsonDecode return a non-Map / throw FormatException. This escapes the SocketException/TimeoutException handlers and surfaces as a raw crash to the caller.

Fix Reuse the defensive _decode() helper (already used by auth/help/coupon/notification services) everywhere; tolerate non-Map / empty bodies.

HIGH No centralized 401 -> token-refresh / retry

Where all services · auth_state.dart:177

Issue The refresh token is only exchanged once at cold start. A mid-session access-token expiry yields inconsistent handling (throw "Session expired", return null, or silent fail) with no automatic refresh-and-retry.

Fix Add an authenticated-request wrapper that, on 401, calls refreshToken(), updates AuthState, retries once, and routes to logout on failure.

MEDIUM Base URL hardcoded & duplicated in 14 files

Where auth_service.dart:8 (+13 siblings) · app_config.dart

Issue The host string is copy-pasted in every service; AppConfig only holds the Razorpay key. No dev/staging/prod switch; a host change means editing 14 files.

Fix Move _base into AppConfig.apiBase and reference it everywhere.

MEDIUM tryRestoreSession keeps an expired access token on network failure

Where auth_state.dart:195-210

Issue When refreshToken throws (offline), it restores from cached tokens and logs in with the old, possibly-expired access token; every subsequent protected call then 401s with no refresh path.

Fix Mark the session "needs refresh" and retry refresh on next foreground / first 401.

MEDIUM Token-rejection detection relies on string-matching the error message

Where `auth_state.dart:188-194`

Issue Whether to clear tokens depends on the message containing "invalid"/"expired"/"blacklisted". A reworded backend message leaves a dead refresh token un-cleared -> broken logged-in-but-401 loop.

Fix Branch on the HTTP status / error code, not free text.

MEDIUM Force-cast IDs in JSON models (as int / as num)

Where `api_review_model.dart:9,88` · `api_followed_partner_model.dart:55`

Issue `json["id"] as int / as num` throws if the backend serializes the id as a string or omits it.

Fix Use `(json["id"] as num?)?.toInt() ?? 0` consistently.

LOW Optimistic-success swallows real failures

Where `wishlist_service.dart:68 (400=success)` · `partner_service.dart (400/404=success)`

Issue A genuine validation 400 is treated as "already saved"/"already followed", so the optimistic UI stays toggled though nothing persisted server-side.

Fix Inspect the error code before assuming idempotent success.

LOW Per-user static caches not reset on logout

Where `notifications_state` · `user_reviews_state.dart:12 (non-scoped prefs key)` · `home_feed_state._loaded` · `saved_events_state`

Issue Static singletons can leak the previous user's unread count / cached reviews into the next account on a shared device.

Fix Scope persisted keys per-user (as `FollowState` does) and reset all static caches in `AuthState.logout()`.

LOW Lost-update race in concurrent loads

Where `saved_events_state.loadFromApi` · `user_reviews_state.loadFromApi`

Issue Unlike `HomeFeedState` (which has a `_loading` guard), these can run concurrently; a slow earlier response can clobber a fresh later one.

Fix Add an in-flight guard or monotonic sequence token.

6. Robustness, Crashes & Edge Cases

HIGH Required casts crash the payment flow

Where `api_booking_model.dart:76-78,81,288` · `review_pay_screen.dart:180-188 (as int)`

Issue `booking_id/razorpay_order_id as String`, `amount as num`, `line-item count as int` cast without fallback. JSON numbers often deserialize as `double`; callers only catch network errors, so a `CastError` surfaces raw during checkout.

Fix Null-coalesce all casts: `(json[x] as String?) ?? ""` ; `(json[x] as num?)?.toDouble() ?? 0`.

HIGH Cover-media as String cast can abort detail parse

Where `api_event_model.dart:220` · `api_venue_model.dart:232` · `api_class_model.dart:196` · `api_program_model.dart:228`

Issue `.map((m) => m["url"] as String)` throws if any cover entry `url/file_url` is null, failing the whole detail-screen parse.

Fix Map to `String?` then `.whereType<String>()` before `firstOrNull`.

HIGH Detail models DateTime.parse without fallback

Where api_event_model.dart:200-225 (+ class/program detail)

Issue DateTime.parse(json["start_datetime"]) throws on a malformed date, where the list model defensively falls back to now().

Fix Use DateTime.tryParse(...) ?? DateTime.now() and null-coalesce required casts.

HIGH accessToken! force-unwrap on session expiry

Where review_sheet.dart:734 · booking_detail_screen.dart:134

Issue If the token is cleared between opening a sheet and tapping submit, accessToken! throws "Null check operator used on null"; the review sheet sticks with _submitting=true.

Fix Read into a nullable local, show an auth error, and bail.

HIGH Empty-list crashes (clamp / modulo / PageView)

Where category_events_screen.dart:48 clamp(0,-1) · special_needs_section.dart:42 index % 0 · trending_card.dart:56 count:0

Issue clamp(0,length-1) throws ArgumentError on an empty list; index % items.length divides by zero; a 0-item PageView + SmoothPageIndicator(count:0) can assert.

Fix Guard isEmpty -> render empty state / SizedBox.shrink() before clamp / modulo / PageView.

HIGH Out-of-range / stale index RangeErrors

Where select_program_batch_screen.dart:39 months[month] · category_classes/events_screen.dart:80/75 _filters[idx]

Issue months[int.tryParse(...)] can index 13; a load-more in flight during a category switch leaves _selectedFilterIndex beyond the new shorter list -> RangeError.

Fix Bounds-check the index (1..12; < list.length) before indexing.

HIGH gradient cast crashes every grid cell on a missing key

Where all_categories_popup.dart:158 · explore_categories_grid.dart:188

Issue category["gradient"] as List<Color> throws TypeError if the key is absent.

Fix (category["gradient"] as List<Color>?) ?? const [..fallback..].

MEDIUM banner_carousel: staticFade + infiniteScroll -> RangeError

Where banner_carousel.dart:105,252

Issue In staticFade mode there is no PageView, so onPageChanged never resets _currentIndex from its _loopBase (lengthx1000) start; widget.events[_currentIndex] would throw. Latent: no caller currently combines both flags.

Fix In staticFade init _currentIndex = 0 regardless, or index with _currentIndex % events.length.

MEDIUM setState after await without mounted guard

Where payment_settings_screen.dart:38-47 · walkthrough_intro_overlay.dart:41

Issue Popping the screen mid-fetch throws "setState() called after dispose".

Fix Add if (!mounted) return; before each post-await setState / callback.

MEDIUM userEmail?.substring(0,1) RangeError on empty string

Where account_settings_screen.dart:57-60

Issue ?. guards null but not "" - an empty first_name/email crashes the profile card.

Fix Guard isEmpty before substring.

MEDIUM Double-tap navigation pushes duplicate routes

Where navbar + checkout CTAs: ticket_booking/seat_reservation/venue_checkout/attendee_details; whole-card taps

Issue onTap calls Navigator.push with no in-flight guard; a fast double-tap stacks identical screens (worst on money flows). review_pay_screen guards correctly.

Fix Shared re-entrancy flag or ModalRoute.of(context)?.isCurrent check before push.

MEDIUM Re-entrant fetch race overwrites with stale results

Where category_events/classes/programs_screen · home_feed_state.dart:24

Issue Overlapping fetches from rapid filter switching can land out of order; HomeFeedState.load also drops a force:true refresh issued during startup.

Fix Monotonic request-id token; discard stale responses; queue forced refresh.

MEDIUM Listing/booking/review services: unguarded decode + a dead branch

Where classes_listing_service.dart:35-54 · events/programs/booking/review_service

Issue jsonDecode+cast inside a try that only catches network errors; classes service has a statically-dead "is List" branch and a jsonDecode(e.toString()) "recovery" that always fails.

Fix Decode as dynamic and branch like programs_listing_service; reuse the _decode helper; wrap review_service in try/catch.

7. UI / UX, Design & Accessibility Findings

HIGH Brand-yellow / blue / background colors drift off AppColors

Where CTAs use 0xFFFFCC00/FFB902/FFD014; ticket/notif UI uses 0xFF2563EB; scaffolds use 0xFFFF2F2F7

Issue AppColors.primary (F5A623), seeAllBlue (0068E7), background (FFF8EE) exist but dozens of screens hardcode competing hexes for the same role - the single biggest consistency gap.

Fix Consolidate to AppColors tokens (CTA gold, accent blue, scaffold bg, text primary/secondary).

HIGH Home feed + 11 sections render DummyData; empty-guards commented out

Where hot_picks_section.dart:21 (+ every section) · home_screen.dart:62,81

Issue isEmpty guards are commented out; sections render today only because dummy lists are non-empty. Once API returns an empty list each section shows a floating divider title over an empty band.

Fix Restore the isEmpty -> SizedBox.shrink() guards before re-enabling live data.

HIGH Failure state masquerades as empty (no retry)

Where upcoming_events_section.dart:61 · category_events_screen.dart:120 (+ siblings)

Issue A network error sets the list to [] and shows the generic empty state - a failed request looks identical to "no results". search_screen does this correctly with a Retry button.

Fix Distinguish error from empty; add a Retry affordance (mirror search_screen).

HIGH Final ticket image is asset-only but receives network URLs

Where booking_confirmed_screen.dart:456

Issue Image.asset(event.imagePath) - after a real booking imagePath is an http URL, so the confirmed ticket always falls to the grey placeholder.

Fix Branch on startsWith("http") -> Image.network, like the detail screens.

HIGH Image.asset calls with no errorBuilder

Where empty_location_widget.dart:22 · home_header.dart:76 · banner_carousel.dart:375 · location_screen.dart:396

Issue A missing/renamed asset renders a broken-image box. Several rely on the "resources- tlb-ui/" path (literal space) which is fragile.

Fix Add errorBuilder fallbacks; verify each asset is declared in pubspec; consider moving under assets/.

HIGH Dead / no-op buttons that look active

Where reminders_screen.dart:242 (Edit) · category_events_screen.dart:247 (See All) · venues_screen.dart:857 (View all) · search fields with no controller

Issue Buttons/fields render as interactive but onTap/onPressed is empty {} - users tap and nothing happens.

Fix Wire the actions or remove the controls / mark them disabled.

HIGH No payment / no-op mock screens shipped alongside real flow

Where payment_screen.dart:90-199 (Future.delayed -> confirmed) · reminders_screen.dart:14 (mock data)

Issue payment_screen lists CRED/Pay-Later tiles that fake a 2s delay then push BookingConfirmed with NO real payment, bypassing the real Razorpay flow; reminders shows static fake events.

Fix Remove/disable the mock screens or wire them to the real flow.

HIGH Systemic sub-44px tap targets + icon-only controls without Semantics

Where navbar (~40) · wishlist_button (36/24) · follow/seat/qty/back buttons (28-36)

Issue Many interactive controls are below the 44px accessibility minimum and announce nothing to screen readers (drawn as GestureDetector+Container / bare IconButton).

Fix Raise tap targets to >=44px; add Semantics labels to icon-only controls.

MEDIUM Fixed widths/heights bypass Responsive (overflow on small screens / large fonts)

Where event_card.dart:18 (260) · weekend_event_card.dart:15 (330) · CTA SizedBox(height:52/54) on checkout screens

Issue Hardcoded card widths and fixed CTA heights can overflow on <360px phones or clip text when OS font scales (currently masked by main.dart clamping scale to 1.0).

Fix Use Responsive.cardWidth/w/h; let CTA height grow with text.

MEDIUM Bottom price+CTA Row can overflow with a long price

Where event_detail_screen.dart:531 · class/program/venue_detail_screen

Issue The price Text/RichText next to the CTA is not Flexible; a long price string overflows the row.

Fix Wrap the price in Flexible/Expanded with ellipsis.

MEDIUM No loading placeholders on network images

Where search/bookings/organizer_card/banner_carousel/listing_image + detail heroes

Issue Image.network has errorBuilder but no loadingBuilder, so covers/avatars pop in or sit blank on slow networks.

Fix Add a loadingBuilder skeleton/shimmer.

MEDIUM White-on-yellow tag text fails WCAG AA contrast**Where** event_detail_screen.dart:317 · class_detail_screen.dart:272 · program_detail_screen.dart:345**Issue** White text on 0xFFFFCC00 chips is low contrast; venue_detail uses dark text correctly.**Fix** Use dark text on the yellow chips consistently.**MEDIUM Placeholder data shipped as real content****Where** all home sections ("3-5 Yrs"/"3.5k reviews") · date_time_selection ("Sun 28 Feb") · fake QR (Random(42)) · gallery demo images**Issue** Hardcoded fallbacks/mocks render identically to real content; the ticket QR is non-scannable.**Fix** Drive from API data; generate the QR from the booking reference; remove demo lists.**MEDIUM booking_confirmed lacks PopScope - back returns into the paid flow****Where** booking_confirmed_screen.dart**Issue** Unlike program/venue confirmed screens (PopScope canPop:false), system-back returns to ReviewPay/"Pay" after payment.**Fix** Add PopScope(canPop:false) and route Home.

8. Prioritized Remediation Plan

#	Theme	Action
1	Crash-safety in money flow	Null-coalesce the required casts in api_booking_model & detail models; fix accessToken! in review_sheet/booking_detail; add empty-list guards (clamp/modulo/PageView).
2	Network resilience	Add a 401->refresh->retry wrapper; route all services through a defensive _decode(); move base URL into AppConfig.
3	Money-flow UX integrity	Add PopScope to booking_confirmed; remove/wire the payment_screen & reminders mocks; debounce navigation on checkout CTAs and the navbar.
4	Live-data readiness	Restore commented-out isEmpty guards in home sections; distinguish error vs empty with Retry; fix asset-only final-ticket image.
5	Design system + a11y	Consolidate gold/blue/bg/text colors into AppColors tokens; raise sub-44px tap targets; add Semantics to icon-only controls; fix white-on-yellow contrast.
6	Polish	Replace shipped placeholder data & fake QR; fix copy typos (Hand-On->Hands-On, Inquire/Enquire); clear large dead/commented code blocks.

9. What Is Solid

- Static analysis is clean: 0 errors, 0 warnings (29 style-only info lints).
- Full automated suite passes: 130/130 widget & unit tests green, including the home smoke test, category screens, review-guard, wishlist toggle, and the new logo SVGs.
- Network timeouts on every call (15-30s); SocketException/TimeoutException/HandshakeException handled with friendly messages.
- Token storage is secure: flutter_secure_storage with encryptedSharedPreferences on Android; tokens never logged.
- Lifecycle hygiene is strong: AnimationController / Timer / PageController / TextEditingController / FocusNode are disposed; most async setState/navigation is mounted-guarded.

- The real payment path (review_pay_screen) is well-built: Razorpay init/dispose, double-submit guard, backend verification, graceful "paid-but-unverified" dialog with reference.
- search_screen is the model: debounced input + full loading/empty/error/no-results states with Retry.
- errorBuilder coverage on raster images is near-universal; optimistic wishlist/follow toggles roll back on failure.
- Font stack is healthy: Poppins genuinely bundled (offline-safe), zero unsupported w300/Light usage.
- Banner carousel core logic is correct: empty-list guard, modulo index wrapping, putIfAbsent palette de-dup, try/catch + fallback color, mounted-guarded band load.

10. Scope & Methodology

Static & automated: flutter analyze (full lib/) and the complete flutter test suite (130 cases) were run on the dev-vishesh branch. Code review: the entire lib/ tree was audited across three dimensions - (1) the data/networking layer (services, providers, api_* models), (2) UI/UX, design tokens, responsiveness, states & accessibility across all 70+ screens / 44 widgets / 11 sections, and (3) lifecycle, null-safety, async correctness, edge-case data & performance.

Not covered (requires a physical device / live backend): real on-device interaction, live API contract verification against the running server, Razorpay sandbox transactions, push-notification delivery, deep-link handling, and platform-specific (iOS/Android) rendering. These are recommended as a follow-up manual QA pass on a device once the Section 7 items land.